

CS 405: Verification of Reactive Systems

Course Code	CS 405
Course Name	Verification of Reactive Systems
L-T-P-C	3-0-0-3
Prerequisites	MFCS, FLAT (recommended)
Intended for	BTech, MTech, MSc, MS/PhD in the area.
Distribution	Discipline elective for BTech CSE and EE, free elective for others
One-time approval	recommended by the 14th BoA (Sep 2016)

Course Contents

- Introduction (1 week)
 - o Introduction and overview of the course
 - o Revision of predicate logic, undecidable problems, proof systems.
 - o Basic approaches to verification: property and equivalence verification, correctness by design
- Modelling of Systems (2 weeks)
 - Formal models for non-terminating reactive systems (hardware/software) : Buechi Automata, Transition Systems, Process Algebras, Petri Nets
 - Dimensions of behavior descriptions :
 - * interleaving vs truly concurrent
 - * linear vs branching time
 - * quantitative vs qualitative descriptions
 - Property classification: regularity, safety, liveness, fairness
 - Equivalence notions:
 - * trace equivalence
 - * testing equivalence
 - * observation equivalence
 - Case studies - equivalence checking
- Property Verification of Reactive Systems (4 weeks)
 - Verification of regular properties
 - Temporal and modal logics for property specification: LTL, CTL, CTL*, HML
 - Verifying properties: principles of model checking, complexity and limits
 - Case studies with model checker SPIN
- Property Verification of Time-Critical and Hybrid Systems (3 weeks)
 - Timed automata (theory, reachability analysis)
 - Case study with model checker UPPAAL
- Correctness by Design (2 week)
 - Stepwise refinement
 - Case study with proof assistant RODIN
- (Remaining week will be used as buffer and for doubt clearing.)

Text Books

1. Christel Baier and Joost-Pieter Katoen, **Principles of Model Checking**, MIT Press, 2008.

References

1. Luca Aceto, Anna Ingolfsdottir, Kim G. Larsen and Jiri Srba, **Reactive Systems - Modelling, Specification and Verification**, Cambridge textbooks, 2007
2. M. Ben-Ari, **Principles of the SPIN Model Checker**, MIT Press, 2008
3. Wan Fokking, **Modelling Distributed Systems**, Springer Verlag, 2007
4. Michael Huth and Mark Ryan, **Logic in Computer Science Modelling and Reasoning about Systems**, Cambridge University Press, 2004
5. Gerald Holzmann, **The SPIN Model Checker - Primer and Reference Manual**, Addison Wesley, 2003
6. Doron Peled, **Software Reliability Methods**, Springer Verlag, 2001